

SAP CPI Project Documentation

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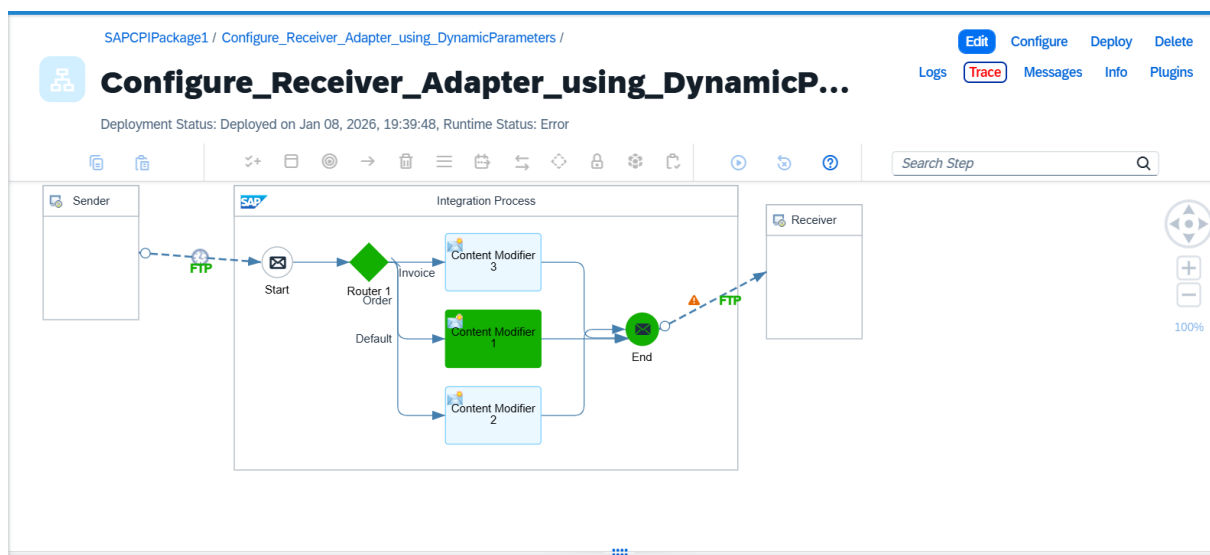
Scenario: Content-Based File Passthrough Interface

“It’s called a Content-Based File integration because the file name and target path are determined dynamically at runtime based on the payload content.”

1.Scenario Overview

This project automates the classification and routing of inbound files received from an upstream system. Files are transferred to SAP Cloud Platform Integration (CPI) via FTP. Based on the file name pattern, SAP CPI intelligently routes the files to different target folders on the FTP server. Files starting with “Order” are routed to the Orders folder, files starting with “Invoice” are routed to the Invoices folder, and all other files are moved to a default Other folder. This solution eliminates manual file segregation and ensures organized downstream processing.

2.Architechture Diagram



3. Step-by-Step Implementation

- Configured FTP Sender Adapter to receive files from the source system
- Implemented Content-Based Routing (CBR) using Regex Expressions
- Dynamically determined the target directory based on business rules:
 - Files starting with “Order” → Orders folder
 - Files starting with “Invoice” → Invoices folder
 - All other files → Other folder
- Configured FTP Receiver Adapter with dynamic directory assignment using message properties.
- Performed end-to-end testing for multiple file name scenarios to ensure accurate routing.

Step1 – Configure the FTP Sender Adapter:

Fetching all files in directory “ALL_FILES”

FTP Adapter (Sender):

The screenshot displays the configuration for the FTP Sender Adapter. The 'Source' tab is active, showing the following settings:

FTP	
General	Source
FILE ACCESS PARAMETERS	
Directory:	ALL_FILES
File Name:	*
CONNECTION PARAMETERS	
Address:	ftp.drivehq.com
Proxy Type:	Internet
Encryption:	Explicit FTPS
Credential Name:	FTP_Connection
Timeout (in ms):	10000
Maximum Reconnect Attempts:	3
Reconnect Delay (in ms):	30000

Step2 – Configure Content -Based Router

The filename of the incoming file will be available in the header parameter CamelFileNameOnly”.

We will route the files based on the prefix of the filename. Using a Regex

expression,

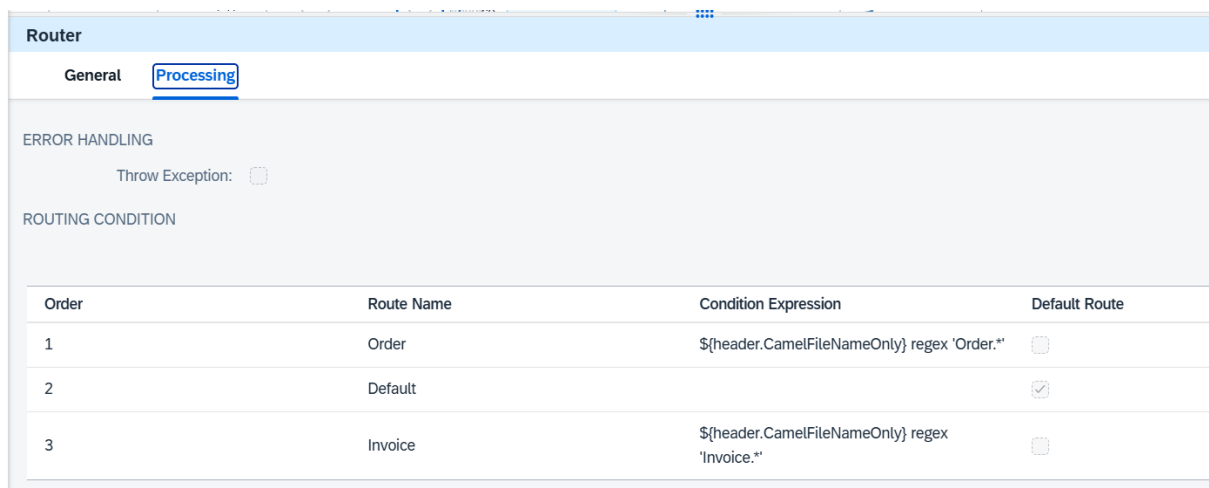
We can find if the filename matches the pattern we are looking for

Regular expression to check if the file name starts with Order. Or Invoice.

`${header.CamelFileNameOnly} regex 'Order.*'`

`${header.CamelFileNameOnly} regex 'Invoice.*'`

Router:



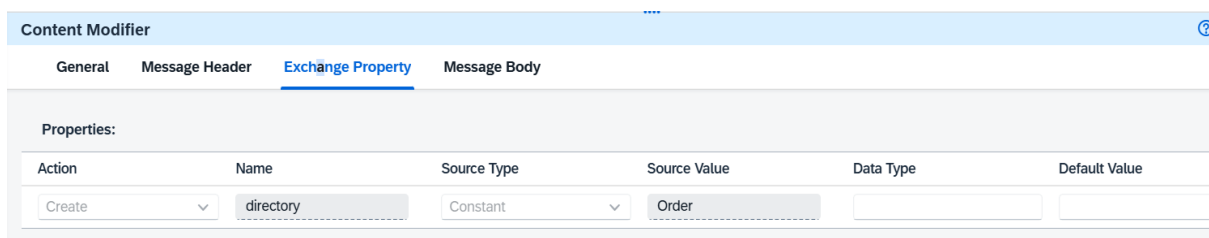
Order	Route Name	Condition Expression	Default Route
1	Order	<code>\${header.CamelFileNameOnly} regex 'Order.*'</code>	☐
2	Default		☒
3	Invoice	<code>\${header.CamelFileNameOnly} regex 'Invoice.*'</code>	☐

Step3 – Make Use of Exchange or Header parameters to set the Directory

Let's make use of content modifiers to determine the directory at runtime.

We will have an exchange property parameter named "directory" to set the value of the directory.

Content Modifier:



Action	Name	Source Type	Source Value	Data Type	Default Value
Create	directory	Constant	Order		

Content Modifier

General Message Header **Exchange Property** Message Body

Properties:

Action	Name	Source Type	Source Value	Data Type	Default Value
Create	directory	Constant	Invoice		

Step4 – Configure Receiver Adapter using Dynamic Parameters

We will make use of Camel Expressions to define the filename dynamically.

`${file:onlyname.noext}` – Camel Simple Expression to get the prefix or the filename of the incoming file without the extension

FTP Adapter (Receiver):

FTP

General **Target** Processing

FILE ACCESS PARAMETERS

Directory: `${property.directory}`

File Name: `${file:onlyname.noext}`

Append Timestamp: ☒

CONNECTION PARAMETERS

Address: `ftp.drivehq.com`

Proxy Type: Internet

Encryption: Explicit FTPS

Credential Name: `FTP_Connect2`

Timeout (in ms): `10000`

Maximum Reconnect Attempts: `3`

Reconnect Delay (in ms): `1000`

6. Input Data:

VAIL_FILES					
Name ▲	Size	Type	Date Modified	Date Created	
InvoiceA.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50	
InvoiceB.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50	
InvoiceC.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50	
Order details1.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50	
Order details2.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50	
Order details3.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50	
POOrders.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50	
PO_Orders.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50	
Sales_Orders.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50	

7. Output Data:

My Storage (\)					
Name ▲	Size	Type	Date Modified	Date Created	
UniqueFiles	578 B	File	2026-01-10 15:59:23	2026-01-10 15:59:23	
DriveHQData	3 KB	Online Backup ...	2026-01-10 12:16:16	2026-01-10 12:16:15	
Express_Orders	1023 B	File Folder	2026-01-10 12:17:27	2026-01-10 12:17:26	
FoodDelivery	662 B	File Folder	2026-01-10 14:00:16	2026-01-10 13:57:41	
FoodDelivery_1	662 B	File Folder	2026-01-10 14:00:16	2026-01-10 13:57:41	
Instamart	311 B	File Folder	2026-01-10 14:04:20	2026-01-10 14:04:19	
Instamart_1	311 B	File Folder	2026-01-10 14:04:20	2026-01-10 14:04:19	
Invoice	26 B	File Folder	2026-01-08 19:34:42	2026-01-08 19:34:32	
My Documents	138 B	File Folder	2026-01-08 19:30:45	2026-01-08 19:30:45	
My Pictures	3,021 KB	File Folder	2026-01-08 19:30:45	2026-01-08 19:30:45	
Online_Orders	1022 B	File Folder	2026-01-10 12:15:26	2026-01-10 12:15:24	
Order	39 B	File Folder	2026-01-08 19:39:57	2026-01-08 19:39:53	
Others	52 B	File Folder	2026-01-08 19:40:02	2026-01-08 19:34:22	
Recycle Bin	3 KB	File Folder	2026-01-10 12:20:17	2026-01-10 12:20:12	
UniqueFiles1	578 B	File Folder	2026-01-10 16:19:42	2026-01-10 16:19:41	
wwwhome	24 KB	File Folder	2026-01-08 19:30:45	2026-01-08 19:30:45	

\\Order				
Name ▲	Size	Type	Date Modified	Date Created
Order details2026010814...	13 B	File	2026-01-08 19:39:53	2026-01-08 19:39:53
OrderC20260108140954	13 B	File	2026-01-08 19:39:55	2026-01-08 19:39:55
OrdersA20260108140956	13 B	File	2026-01-08 19:39:56	2026-01-08 19:39:57

\\Invoice				
Name ▲	Size	Type	Date Modified	Date Created
InvoiceB20260108140431	13 B	File	2026-01-08 19:34:32	2026-01-08 19:34:32
InvoiceC20260108140441	13 B	File	2026-01-08 19:34:41	2026-01-08 19:34:42

\\Others				
Name ▲	Size	Type	Date Modified	Date Created
InvoiceA20260108140421	13 B	File	2026-01-08 19:34:23	2026-01-08 19:34:23
POOrders20260108141000	13 B	File	2026-01-08 19:40:00	2026-01-08 19:40:01
PO_Orders202601081409...	13 B	File	2026-01-08 19:39:58	2026-01-08 19:39:59
Sales_Orders2026010814...	13 B	File	2026-01-08 19:40:02	2026-01-08 19:40:02

8. Iflow Package:



Configure_Receiver_Adapter_using_DynamicParameters.zip